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Roll No

CE-602 (GS)

B.E. VI Semester Examination, June 2020

Grading System (GS)

Water Resources and Irrigation Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Describe flowing artesian well and non flowing artesian well with neat sketch.
2. What do you understand by flood routing? Write the basic equation of flood routing.
3. a) Discuss in brief various methods of surface irrigation.
b) Derive a relationship between duty and delta.
4. A tube well is used to irrigate rabi crop to cover an area of 0.2 hectare. The discharge from the well is $0.04\text{m}^3/\text{sec}$. The average depth of flow of the crop is expected to be 10cm with an infiltration of 4cm/h, find the time required to irrigate and the maximum area that can be irrigated.
5. Define the following terms:
 - i) Aquifer
 - ii) Aquiclude
 - iii) Specific Yield
 - iv) Perched Aquifer
6. What do you understand by mass rainfall curves and hydrograph.
7. Design an irrigation channel in alluvial soil according to Lacey's silt theory for the following data :
Full supply discharge $50\text{m}^3/\text{sec}$., Lacey's silt factor 1.0 and side slope of channel $\frac{1}{2}\text{H} : 1\text{V}$.
8. Briefly explain any three of the following:
 - i) Salt efflorescence
 - ii) Specific capacity
 - iii) Well irrigation
 - iv) Spillways
 - v) S-curves Hydrograph
